

Fari-Class Indoor Platform Power Budget & Mission Time Estimation

Design Philosophy

The Fari-class indoor platform prioritizes **ultra-low acoustic noise (completely fanless, passive cooling)**, a highly optimized compute envelope, and high-efficiency indoor localization. This budget is based on the same **4S2P 18650 battery pack at 80% usable capacity (80.64 Wh)** to provide a direct baseline comparison with the Aido-class platform.

1. Battery Configuration Baseline (4S2P 18650 Pack)

Parameter	Value	Unit	Notes / Formulas
Battery Chemistry	Lithium-Ion (18650)	-	Standard high-density cells
Pack Nominal Voltage	14.4	V	4 series cells * 3.6V nominal
Pack Total Capacity	7.0	Ah	2 parallel cells * 3.5Ah nominal
Total Nominal Energy	100.8	Wh	= 14.4 V * 7.0 Ah
Total Usable Energy	80.64	Wh	= 100.8 Wh * 80% usable capacity factor

2. Component Power Consumption Breakdown

To guarantee completely silent operation, the compute profile is restricted to a low-TDP setting (e.g., Jetson Orin Nano capped at 7W or an optimized single-board computer). Drive motors are downscaled for smooth, low-friction indoor floors, and LTE comms are disabled in favor of local Wi-Fi.

Component	Nominal Power (W)	Peak Power (W)	Duty Cycle (%)	Average Power (W)
Compute (Fanless / Low-TDP Profile)	7.0	10.0	100%	7.00

Component	Nominal Power (W)	Peak Power (W)	Duty Cycle (%)	Average Power (W)
Drive Motors (Quiet Indoor Actuators)	8.0	20.0	40%	3.20
Sensor Stack (Low-power LiDAR + IMU)	4.0	6.0	100%	4.00
Communications (Wi-Fi Only, LTE Disabled)	2.5	5.0	50%	1.25
Payload Allowance (Indoor-specific)	5.0	10.0	30%	1.50

3. Power Budget Summary

Metric	Value	Unit	Calculation / Description
Continuous Power Draw	11.00	W	Compute (7W) + Sensors (4W)
Average Dynamic Power Draw	5.95	W	Motors (3.2W) + Comms (1.25W) + Payload (1.5W)
Total Average System Power	16.95	W	Continuous (11W) + Average Dynamic (5.95W)
Total Peak System Power	51.00	W	Sum of all peak power limits (Critical for passive heatsink sizing)

4. Operational Mission Time Estimation

Parameter	Value	Unit	Formula
Total Usable Energy Available	80.64	Wh	From Battery Configuration

Parameter	Value	Unit	Formula
Total Average System Power Draw	16.95	W	From Power Budget Summary
Estimated Mission Time (Hours)	4.76	Hours	$= 80.64 \text{ Wh} / 16.95 \text{ W}$
Estimated Mission Time (Minutes)	285.4	Minutes	$= 4.76 * 60$
Final Formatted Runtime	4 Hours 45 Minutes	-	Continuous indoor operation at 80% capacity